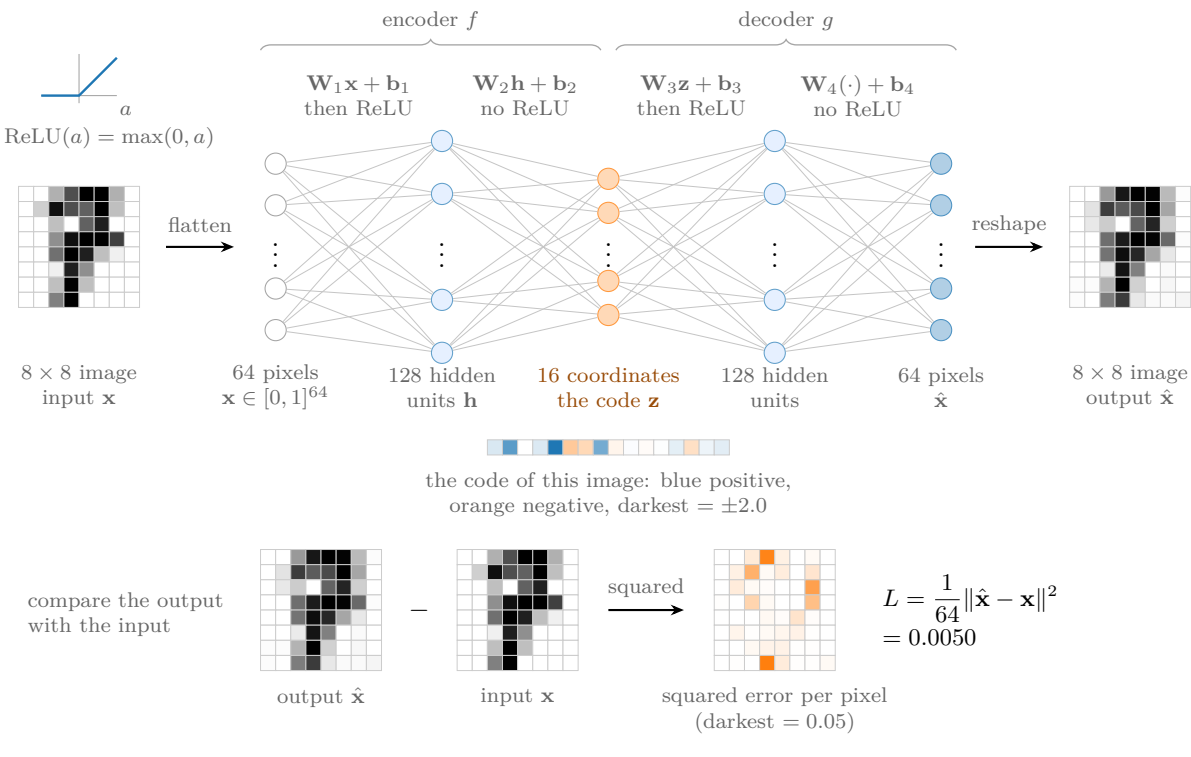

encoder f
decoder g
a
$\mathrm{ReLU}(a) = \max(0, a)$
$\mathbf{W}_1\mathbf{x} + \mathbf{b}_1$ then ReLU
$\mathbf{W}_2\mathbf{h} + \mathbf{b}_2$ no ReLU
$\mathbf{W}_3\mathbf{z} + \mathbf{b}_3$ then ReLU
$\mathbf{W}_4(\cdot) + \mathbf{b}_4$ no ReLU
flatten
reshape
8×8 image input $\mathbf{x}$
64 pixels $\mathbf{x} \in [0, 1]^{64}$
128 hidden units $\mathbf{h}$
16 coordinates the code $\mathbf{z}$
128 hidden units
64 pixels $\hat{\mathbf{x}}$
8×8 image output $\hat{\mathbf{x}}$
the code of this image: blue positive, orange negative, darkest $= \pm 2.0$
compare the output with the input
output $\hat{\mathbf{x}}$
input $\mathbf{x}$
squared
squared error per pixel (darkest $= 0.05$)
$L = \frac{1}{64}\|\hat{\mathbf{x}} - \mathbf{x}\|^2$
$= 0.0050$